

1.

$$x_{n+1} = 2x_n - 2(x_n)^2$$

$$\bar{x} = 2\bar{x} - 2\bar{x}^2$$

$$2\bar{x}^2 - \bar{x} = 0$$

$$\bar{x}(2\bar{x} - 1) = 0$$

$$\text{FIXED POINTS: } \bar{x} = 0 \wedge \bar{x} = \frac{1}{2}$$

STABILITY

$$F(x) = 2x - 2x^2$$

$$F_x(x) = 2 - 4x$$

$$F_x(0) = 2 \Rightarrow |F_x(0)| > 1 \text{ unstable}$$

$$F_x\left(\frac{1}{2}\right) = 2 - 2 = 0 \Rightarrow |F_x\left(\frac{1}{2}\right)| < 1 \text{ stable}$$

2.

$$x_{n+1} = \frac{1}{2}x_n - (x_n)^2 + \frac{1}{2}$$

$$\bar{x} = \frac{1}{2}\bar{x} - \bar{x}^2 + \frac{1}{2}$$

$$\bar{x}^2 + \frac{1}{2}\bar{x} - \frac{1}{2} = 0$$

$$(\bar{x} + 1)(\bar{x} - \frac{1}{2}) = 0$$

$$\text{FIXED POINTS: } \bar{x} = -1 \wedge \bar{x} = \frac{1}{2}$$

STABILITY

$$F(x) = \frac{1}{2}x - x^2 + \frac{1}{2}$$

$$F_x(x) = \frac{1}{2} - 2x$$

$$F_x(-1) = \frac{5}{2} \Rightarrow |F_x(-1)| > 1 \text{ unstable}$$

$$F_x\left(\frac{1}{2}\right) = \left(-\frac{1}{2}\right) \Rightarrow |F_x\left(\frac{1}{2}\right)| < 1 \text{ stable}$$

3.

$$x_{n+1} = (x_n)^2 - x_n - 2$$

$$\bar{x} = \bar{x} - \bar{x} - 2$$

$$\bar{x}^2 - 2\bar{x} - 2 = 0$$

$$\bar{x} = \frac{2 \pm \sqrt{4+8}}{2} = 1 \pm \sqrt{3}$$

$$\text{FIXED POINTS: } \bar{x} = 1 + \sqrt{3} \wedge \bar{x} = 1 - \sqrt{3}$$

STABILITY

$$F(x) = x^2 - x - 2$$

$$F_x(x) = 2x - 1$$

$$F_x(1 + \sqrt{3}) = 1 + 2\sqrt{3} \Rightarrow |F_x(1 + \sqrt{3})| > 1 \text{ unstable}$$

$$F_x(1 - \sqrt{3}) = 1 - 2\sqrt{3} \Rightarrow |F_x(1 - \sqrt{3})| > 1 \text{ unstable}$$

4.

$$x_{n+1} = x_n (x_n - 1)(x_n + 1)$$

$$\bar{x} = \bar{x} (\bar{x} - 1)(\bar{x} + 1)$$

$$\bar{x} [(\bar{x} - 1)(\bar{x} + 1) - 1] = 0$$

$$\bar{x} (x^2 - 2) = 0$$

$$\bar{x} (x + \sqrt{2})(x - \sqrt{2}) = 0$$

$$\text{FIXED POINTS: } \bar{x} = 0 \wedge \bar{x} = -\sqrt{2} \wedge \bar{x} = \sqrt{2}$$

STABILITY

$$F(x) = x(x-1)(x+1) = x(x^2 - 1) = x^3 - x$$

$$F_x(x) = 3x^2 - 1$$

$$F_x(0) = -1 \Rightarrow |F_x(0)| = 1$$

we cannot tell with linear stability alone.

$$F_x(\sqrt{2}) = 5 \Rightarrow |F_x(\sqrt{2})| > 1 \text{ unstable}$$

$$F_x(-\sqrt{2}) = 5 \Rightarrow |F_x(-\sqrt{2})| > 1 \text{ unstable}$$

LOGISTIC MAP

$$x_{n+1} = a x_n (2 - x_n)$$

$$0 \leq x \leq 2 \quad a > 0$$

FIXED POINTS

$$\bar{x} = a \bar{x} (2 - \bar{x})$$

$$\bar{x} = 2a\bar{x} - a\bar{x}^2$$

$$a\bar{x}^2 + (1 - 2a)\bar{x} = 0$$

$$\bar{x} (a\bar{x} + (1 - 2a)) = 0$$

$$\boxed{\bar{x} = 0 \quad \wedge \quad \bar{x} = \frac{2a-1}{a}}$$

EXISTENCE CONDITION

$$0 \leq \frac{2a-1}{a} \leq 2$$

$$0 \leq 2a-1 \leq 2a \quad \begin{cases} 2a-1 \geq 0 & a \geq \frac{1}{2} \\ 2a-1 \leq 2a & \forall a \in \mathbb{R} \end{cases}$$

$$\boxed{a \geq \frac{1}{2}}$$

STABILITY

$$f(x) = a x (2 - x) = 2ax - ax^2$$

$$f'(x) = 2a - 2ax$$

$$f'(0) = 2a \Rightarrow \bar{x} = 0 \text{ stable if } a < \frac{1}{2}$$

$$f'\left(\frac{2a-1}{a}\right) = 2a - 2a\left(2 - \frac{1}{a}\right) = 2a - 4a + 2 = -2a + 2$$

$$\bar{x} = 2 - \frac{1}{a} \text{ stable if } |-2a + 2| < 1$$

which means

$$-1 < -2a + 2 < 1$$

$$-1 - 2 < -2a < 1 - 2$$

$$-3 < -2a < -1$$

$$\frac{3}{2} > a > \frac{1}{2}$$

$$\text{For } \frac{1}{2} < a < \frac{3}{2} \quad \bar{x} = 2 - \frac{1}{a} \text{ is stable.}$$

PERIOD - 2 POINTS

$\bar{x}$ is a period-2 point if $\bar{x} = f(f(\bar{x}))$ and $\bar{x} \neq f(\bar{x})$

$$f(x) = a x(2-x) = 2ax - ax^2$$

$$f(f(x)) = 2a(2ax - ax^2) - a(2ax - ax^2)^2 =$$

$$= 4a^2x - 2a^2x^2 - 4a^3x^2 + 4a^3x^3 - a^3x^4$$

We need to solve the equation

$$a^3x^4 - 4a^3x^3 + 4a^3x^2 + 2a^2x^2 - 4a^2x + x = 0$$

$$x(a^3x^3 - 4a^3x^2 + (4a^3 + 2a^2)x + (1 - 4a^2)) = 0$$

$$x\left(x^3 - 4x^2 + \left(4 + \frac{2}{a}\right)x + \left(\frac{1}{a^3} - \frac{4}{a}\right)\right) = 0$$

We know that $x = 2 - \frac{1}{a}$ is a factor

	1	-4	$\left(4 + \frac{2}{a}\right)$	$\left(\frac{1}{a^3} - \frac{4}{a}\right)$
$2 - \frac{1}{a}$		$2 - \frac{1}{a}$	$-4 + \frac{2}{a} \cdot \frac{2}{a} + \frac{1}{a^2}$	$\frac{4}{a} - \frac{2}{a^2} + \frac{2}{a^2} - \frac{1}{a^3}$
	1	$-2 - \frac{1}{a}$	$\frac{2}{a} + \frac{1}{a^2}$	0

✓

$$x\left(x - \left(2 - \frac{1}{a}\right)\right)\left(x^2 - \left(2 + \frac{1}{a}\right)x + \left(\frac{2}{a} + \frac{1}{a^2}\right)\right) = 0$$

$$x\left(x - \left(2 - \frac{1}{a}\right)\right)\left(a^2x^2 - (2a^2 + a)x + (2a + 1)\right) = 0$$

$$x = \frac{(2a^2 + a) \pm \sqrt{(2a^2 + a)^2 - 4a^2(2a + 1)}}{2a^2} = \frac{2a + 1 \pm \sqrt{4a^2 - 4a - 3}}{2a}$$

they exist if

$$4a^2 - 4a - 3 \geq 0$$

roots

$$a = \frac{4 \pm \sqrt{16+48}}{8} = \frac{4 \pm 8}{8} = \begin{cases} \frac{3}{2} \\ -\frac{1}{2} \end{cases}$$

$$a \leq -\frac{1}{2} \quad \vee \quad a \geq \frac{3}{2}$$

since $a > 0$

$$\boxed{a \geq \frac{3}{2}}$$

STABILITY OF PERIOD-2 POINTS

A period-2 point $\bar{x}$ is an attractor if $\left| \frac{d}{dx} (f(f(x))) \right|_{x=\bar{x}} < 1$

$$f(f(x)) = 4a^2x - 2a^2x^2 - 4a^3x^2 + 4a^3x^3 - a^3x^4$$

$$\frac{d}{dx} (f(f(x))) = 4a^2 - 4a^2x - 8a^3x + 12a^3x^2 - 4a^3x^3$$

$$\frac{d}{dx} \left(f \left(f \left(\frac{2a+1 \pm \sqrt{4a^2-4a-3}}{2a} \right) \right) \right) = -4a^2 + 4a + 4$$

the period-2 points are stable if $|-4a^2 + 4a + 4| < 1$

$$-1 < -4a^2 + 4a + 4 < 1$$

$$-4a^2 + 4a + 4 > -1$$

$$-4a^2 + 4a + 4 < 1$$

$$-4a^2 + 4a + 5 > 0$$

$$-4a^2 + 4a + 3 < 0$$

$$a^2 - a - \frac{5}{4} < 0$$

$$a^2 - a - \frac{3}{4} > 0$$

roots

$$a = \frac{1 \pm \sqrt{1+5}}{2} = \frac{1 \pm \sqrt{6}}{2}$$

$$a = \frac{1 \pm \sqrt{1+3}}{2} = \frac{1 \pm 2}{2} = \begin{cases} \frac{3}{2} \\ -\frac{1}{2} \end{cases}$$

$$\frac{1-\sqrt{6}}{2} < a < \frac{1+\sqrt{6}}{2}$$

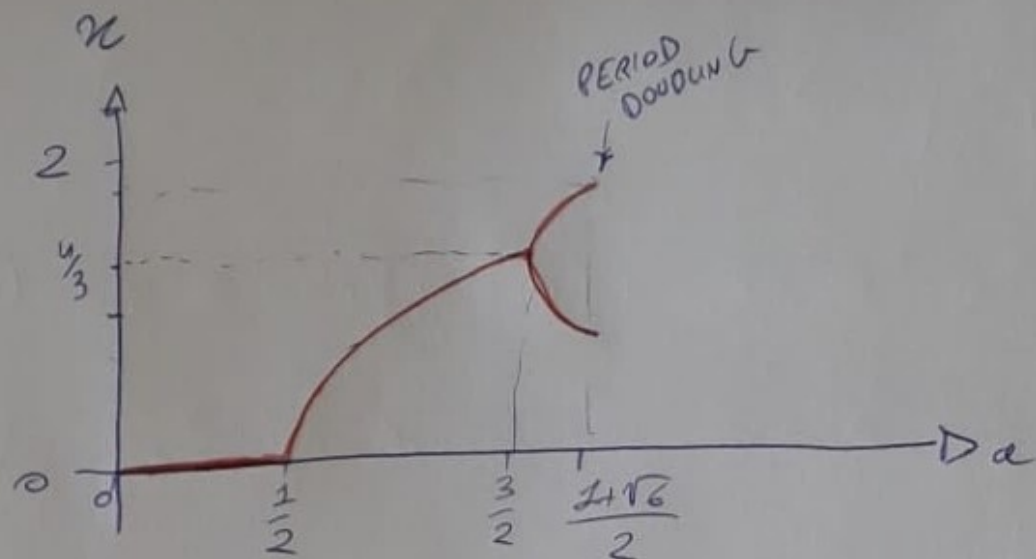
$$a < -\frac{1}{2} \quad \vee \quad a > \frac{3}{2}$$

$$0 < a < \frac{1+\sqrt{6}}{2}$$

$$a > \frac{3}{2}$$

$$\boxed{\frac{3}{2} < a < \frac{1+\sqrt{6}}{2}}$$

$$\frac{1+\sqrt{6}}{2} \approx 1.725$$



$$x = 2 - \frac{1}{a} \quad \text{for } a = \frac{3}{2} \quad x \approx 1.333 = \frac{4}{3}$$

$$\frac{2a + 1 \pm \sqrt{4a^2 - 4a - 5}}{2a} = \begin{cases} \approx 1.7 \\ \approx 0.88 \end{cases} \quad \left| a = \frac{1+\sqrt{6}}{2} \right.$$